MANIPAL INSTITUTE OF TECHNOLOGY
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Mini Project Problem Statement
of
MACHINE LEARNING (CSE 3125)

AI-Based Summarization and Clause Classification of Indian Legal Documents Using NLP and Transformer Models

SUBMITTED
BY

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Initial Mini-Project Problem Statement

AI-Based Summarization and Clause Classification of Indian Legal Documents Using NLP and Transformer Models

Introduction and Real-World Problem

Indian Supreme Court judgments can be lengthy, and a reader must still identify the issues, reasoning and final order. Commercial contracts present a different reading problem: important obligations may be distributed across many passages and a passage can express several clause types. This mini-project builds a local tool that places its output beside the source text so a reader can verify it. The first task summarizes Indian judgments using the public IN-Abs collection of judgment–headnote pairs. The second locates eight types of clauses using CUAD, a public expert-annotated collection of English commercial contracts from United States filings. CUAD is therefore a method benchmark, not evidence that the classifier has been validated on Indian contracts. We compare an extractive summary with a fine-tuned small generator, and a TF-IDF clause classifier with a fine-tuned legal-language encoder. Original documents are split before passages are made; duplicate and annotation-offset checks limit avoidable leakage or label errors. Evaluation uses held-out ROUGE and multi-label precision, recall and F1, with visible examples of omissions and false detections. The proposed use is research assistance under human review, not a legal conclusion or replacement for reading the judgment or agreement.

Keywords

clause classification; Indian judgments; legal NLP; multi-label learning; summarization.

Objectives

1. Summarize Indian Supreme Court judgments from verified judgment–headnote pairs, comparing an extractive baseline and a fine-tuned generator.
2. Convert eight verified CUAD clause spans into multi-label passages and compare a lexical classifier with a fine-tuned legal encoder.
3. Report held-out task-specific scores and source-linked errors in a local, exportable interface.

Data Sources and Scope

IN-Abs: Shukla et al., Zenodo record 7152317, DOI 10.5281/zenodo.7152317, CC BY 4.0; English Indian Supreme Court judgment–headnote pairs. CUAD v1: The Atticus Project, https://www.atticusprojectai.org/cuad, CC BY 4.0; English SEC commercial contracts with expert QA spans. CUAD is not an Indian contract benchmark. Operational provenance, checksums and label conversion are supplied in docs/DATA_CARD.md and docs/SOURCE_RECORD.md.

Three-Student Team and Guide

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